

Bright Coffee Shop: CEO Revenue Growth Initiative

Primary Goal: Increase overall revenue and product performance

Bright Coffee Shop Sales Analysis

DUE DATE: 31 August 2026

Introduction

You have been provided with a dataset titled "Bright Coffee Shop Sales", which captures daily transactional information from a coffee shop. The business has recently appointed a new CEO whose mission is to grow the company's revenue and improve product performance. Your role, as a Junior Data Analyst, is to extract actionable insights from historical data and prepare a presentation to assist the CEO in decision-making.

Objective

Use your analytics, SQL, and data visualization skills to help Bright Coffee Shop understand:

- Which products generate the most revenue
- What time of day the store performs best
- Sales trends across products and time intervals
- Recommendations for improving sales performance

Tools Allowed

You may use any tool at your disposal, including but not limited to:

Project Planning	Coding Platforms	Data Visualisation	Presentation & Reporting
Miro	Microsoft SQL Server	Microsoft Excel	Microsoft PowerPoint
Figma	Databricks	Power BI	Cans
	Google BigQuery	Tableau	Miro
	MySQL Workbench	Google Sheets	Figma
	DBViewer		

Tasks

Tasks 1: Planning & Architecture (Miro)

1. Design a Data Flow & Architecture Diagram that outlines:

- Where the data comes from (source)
- How it is processed (ETL pipeline)
- Where it is stored (Databricks)
- How it is analysed and presented

2. List the Key Insights that the team should deliver, such as:

- Sales by product category and time intervals
- High-performing and low-performing products
- Total revenue calculations

3. Outline the calculations to be performed:

- Total Amount = unit_price * transaction_qty
- Grouping by 30-minute time intervals
- Total units sold by product type or detail

You can add on the above, this is just to give you an idea of what is expected from the Miro planning.

Tasks 2: Data Processing in Databricks

1. Convert the provided Excel data to CSV

2. Load the CSV into Databricks

3. Perform data transformations:

- Create a new column: "transaction_time_bucket" to group transactions into 30-minute intervals (Or it can be 5 hour intervals)
- Cast "unit_price" property (some entries use commas, e.g., "3.1" should be converted to 3.1)
- Compute "total_amount" = unit_price * transaction_qty
- Use SQL to group by product types, time buckets, etc.

You can add on the above, this is just to give you an idea of what is expected from the Data Processing.

Tasks 3: Data Analysis in Excel

After processing the data in Databricks:

1. Export the processed table to Microsoft Excel or your visualization tool

2. Create dashboards or pivot tables showing:
- Total revenue per product type
 - Peak time intervals for sales
 - Quantity of items sold by product category
 - Best-selling product types or details
 - Use charts and graphs to make the story visually appealing

Tasks 4: Presentation to the CEO

1. Key insights from your analysis (supported by visuals) - Create a separate document for your Presentation

2. Recommendations:
- Marketing campaigns during slow time slots
 - Stock more of the best-selling items
 - Promote underperforming products

3. Next Steps:
- Automate daily sales reporting
 - Track sales performance across multiple locations in the future
 - Implement loyalty programs based on peak customer time slots

Submission Guidelines (GitHub)

Submit the following items:

- Miro Plan/Diagram
- Processed Dataset on a Spreadsheet (Microsoft Excel or Google Sheets) with Pivot tables & Charts
- PowerPoint Presentation
- SQL file with your code

DATA SOURCE

Where data comes from

POS system / sales transactions columns

. transaction_id
. transaction_date
. transaction_time
. transaction_qty
. store_location
. store_id
. product_id
. unit_price
. product_category
. product_type
. product_detail

PROCESS (ETL)

How it is processed

1. clean the data - Databricks

2. fix missing values

3. Add new useful columns:

. total_amount
. 3_hour_time_bucket
. time_bucket
. hour_of_day
. day_of_week
. total_amount_band
. day_classification
. month_name

STORE

Where it is stored

Databricks as

1. Raw data
2. Cleaned data
3. Ready for analysis table as Excel / CSV

ANALYSIS & PRESENTATION

How it is used

1. Powe BI dashboard
2. Excel dashboard
3. Lovable dashboard
4. Data studio dashboard
5. Databricks dashboard
- 6 PowerPoint final presentation

KEY INSIGHTS TO DELIVER

1. Sales by category and time
2. High and low performing products
3. Total revenue by store
4. Best hours and day to sell
5. Areas for improvement
6. Recommendations